CUSTOMER SENTIMENT AND INTENT ANALYSIS PLATFORM FOR BUSINESS INTELLIGENCE

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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Course Name & Code

DSA0307 – NATURAL LANGUAGE PROCESSING

Under the Supervision of

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DECLARATION

I declare that the report entitled **“CUSTOMER SENTIMENT AND INTENT ANALYSIS PLATFORM FOR BUSINESS INTELLIGENCE”,** is a unique and original work, is submitted by me for the degree of Bachelor of Engineering. This work, a record of the capstone project for the course **DSA0307 – NATURAL LANGUAGE PROCESSING** was carried out by me under the guidance of **Dr. K. Anbazhagan** and will not form the basis for the award of any degree or diploma in this or any other University or other similar institution of higher learning**.**

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BONAFIDE CERTIFICATE

Certified that this capstone project reports on **"CUSTOMER SENTIMENT AND INTENT ANALYSIS PLATFORM FOR BUSINESS INTELLIGENCE",** is the Bonafide work of **C. Jaswanth(192424334),M. Phanith Chandu(192425238),S. Mahammed Nayeem(192425294),B. Ganga Niranjan (192424041),S. Tushar Krishna Sai (192424406)** who carried out the capstone project work under my supervision for the course **DSA0307 – NATURAL LANGUAGE PROCESSING**

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

The **Customer Sentiment and Intent Analysis Platform for Business Intelligence** is an AI-based system designed to analyze customer feedback, reviews, surveys, and online comments to understand customer opinions and intentions. The platform uses **Natural Language Processing (NLP)** and **Machine Learning** techniques to classify customer sentiment as positive, negative, or neutral and identify customer intent such as complaints, inquiries, requests, feedback, or purchase interest. The analyzed data is presented through an interactive **Business Intelligence dashboard** containing sentiment trends, intent distribution, customer insights, and performance indicators. This helps businesses quickly identify customer concerns, understand changing preferences, improve products and services, and make data-driven decisions. By converting large volumes of unstructured customer text into meaningful insights, the proposed platform supports **customer satisfaction, business growth, and strategic decision-making**.

The system collects and preprocesses customer-generated text by performing operations such as **text cleaning, tokenization, stop-word removal, and normalization**. NLP and machine learning models are then used to determine the customer's sentiment as **positive, negative, or neutral**. In addition to sentiment, the platform identifies customer intent, including **product inquiries, complaints, purchase interest, feedback, support requests, cancellations, and appreciation**. This enables businesses to understand not only *what customers are saying* but also *what they expect or want from the organization*.

The analyzed information is integrated into a **Business Intelligence dashboard** that provides visual reports, charts, sentiment trends, intent distributions, customer satisfaction indicators, and frequently occurring issues. Managers can use these insights to identify negative customer experiences, recognize emerging problems, measure service quality, and understand changing customer preferences. The platform can also generate **actionable recommendations**, helping organizations respond quickly to customer needs and improve their products and services.

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Sincerely,

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CHAPTER 1
INTRODUCTION

1.1 Background Information

In today's digitally connected business environment, customers interact with organizations through online reviews, social media, surveys, emails, chat applications, and customer support platforms. These interactions generate a large amount of valuable customer feedback. However, manually analyzing such feedback is time-consuming and may not provide accurate information about customer emotions and expectations. **Sentiment Analysis** and **Intent Analysis** provide an effective way to automatically understand customer opinions and requirements from textual data. This project presents a **Customer Sentiment and Intent Analysis Platform for Business Intelligence**, which uses Natural Language Processing (NLP) and Machine Learning techniques to analyze customer feedback. The system identifies whether the feedback is positive, negative, or neutral and determines the customer's intent, such as complaint, inquiry, purchase interest, support request, or feedback. The analyzed information is presented through Business Intelligence dashboards, helping organizations identify customer concerns, understand satisfaction trends, and make better data-driven decisions.

1.2 Project Objectives

- To develop an intelligent platform for analyzing customer feedback using NLP and Machine Learning techniques.
- To classify customer feedback into positive, negative, and neutral sentiments.
- To identify customer intentions such as complaints, inquiries, requests, purchase interest, and feedback.
- To preprocess and analyze large amounts of unstructured customer text efficiently.
- To provide Business Intelligence dashboards for visualizing sentiment and intent analysis results.
- To identify common customer problems and changing customer preferences.
- To support organizations in making faster and data-driven business decisions.

- To improve customer satisfaction by providing meaningful insights into customer needs and expectations.

1.3 Significance

The proposed platform helps businesses understand customer opinions and requirements without depending completely on manual feedback analysis. It converts unstructured customer comments into meaningful information that can be easily interpreted by managers and business analysts. Sentiment analysis helps identify customer satisfaction levels, while intent analysis helps determine the actions expected by customers. The Business Intelligence dashboard provides clear visualizations of customer trends, complaints, and preferences. This can help organizations improve their products and services, enhance customer support, identify potential problems at an early stage, and improve overall customer experience.

1.4 Scope

The system focuses on analyzing textual customer feedback collected from sources such as reviews, surveys, social media comments, emails, and customer support messages. It covers **data collection, text preprocessing, sentiment classification, intent detection, data storage, visualization, and business insight generation**. The platform can be applied in industries such as e-commerce, banking, retail, telecommunications, hospitality, and customer service. The system can also be extended in the future to support multilingual feedback, real-time analysis, emotion detection, predictive analytics, and automated business recommendations.

1.5 Methodology Overview

The proposed system follows a systematic workflow for converting customer feedback into useful business intelligence. Initially, customer feedback is collected and stored in a structured format. The text is then cleaned and preprocessed using NLP techniques such as tokenization, stop-word removal, and normalization. Relevant features are extracted from the processed text and provided to Machine Learning or NLP models for **sentiment and intent classification**. The analysis results are stored for further processing and displayed through interactive Business Intelligence dashboards. The dashboard provides charts, graphs, sentiment trends, intent distributions, and frequently occurring customer issues.

CHAPTER 2
PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Businesses receive a large volume of customer feedback through online reviews, surveys, social media, emails, chat messages, and customer support platforms. This feedback contains important information about customer satisfaction, opinions, complaints, expectations, and purchasing intentions. However, analyzing such a large amount of unstructured textual data manually is time-consuming and difficult. Traditional methods mainly depend on ratings, surveys, and manual review, which may not accurately capture the emotions and actual intentions of customers. As a result, businesses may fail to identify common problems, respond quickly to complaints, or understand changing customer preferences. The proposed **Customer Sentiment and Intent Analysis Platform for Business Intelligence** addresses this problem by using NLP and Machine Learning techniques to automatically analyze customer feedback, identify sentiment and intent, and present the results through Business Intelligence dashboards.

2.2 Evidence of the Problem

The increasing use of digital platforms has resulted in a continuous growth of customer-generated content. Organizations may receive hundreds or thousands of reviews and messages, making manual analysis inefficient. Different customers may express similar opinions using different words, while some messages may contain both positive and negative statements. Simple star ratings also cannot explain the reason behind a customer's satisfaction or dissatisfaction. In addition, customer messages may contain different intentions, such as requesting information, reporting a problem, asking for a refund, or showing interest in purchasing a product. These challenges demonstrate the need for an automated system that can analyze customer feedback efficiently and provide meaningful insights.

2.3 Stakeholders

The major stakeholders of the proposed system include:

- **Business Managers:** Use customer insights to make strategic decisions and improve business performance.
- **Customer Service Teams:** Identify complaints and customer requirements for faster response.
- **Marketing Teams:** Analyze customer opinions and understand market trends.
- **Product Managers:** Identify product-related issues and customer preferences.
- **Business Analysts:** Study sentiment and intent trends using dashboards and reports.
- **Customers:** Benefit from improved products, services, and customer support.

2.4 Supporting Data/Research

Customer feedback is an important source of information for understanding market expectations and service quality. Sentiment analysis can classify textual feedback into positive, negative, and neutral categories, while intent classification helps identify the purpose of customer communication. NLP techniques allow large volumes of text to be processed automatically, reducing the effort involved in manual analysis. Business Intelligence tools further help organizations convert analyzed data into dashboards, reports, and visual insights. Combining these technologies provides an effective approach for transforming raw customer feedback into actionable information that supports data-driven decision-making.

CHAPTER 3
SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development & Design Process

The development of the **Customer Sentiment and Intent Analysis Platform for Business Intelligence** follows a systematic process to convert raw customer feedback into meaningful business insights. Initially, customer feedback is collected from sources such as reviews, surveys, social media, emails, and customer support messages. The collected data is cleaned and preprocessed using Natural Language Processing techniques such as tokenization, stop-word removal, text normalization, and removal of unnecessary characters. After preprocessing, relevant features are extracted from the text and passed to suitable Machine Learning or NLP models. The system then performs sentiment analysis to classify feedback as positive, negative, or neutral and intent analysis to identify customer requirements such as complaints, inquiries, purchase interest, support requests, and feedback. The results are stored in a structured database and presented through a Business Intelligence dashboard using charts, graphs, and reports. This development process ensures that raw customer feedback is transformed into understandable and actionable information.

3.2 Tools & Technologies Used

The proposed system uses a combination of programming languages, NLP libraries, Machine Learning frameworks, databases, and visualization tools.

- **Python:** Used as the primary programming language for developing the analysis system.
- **Pandas:** Used for handling, cleaning, and analyzing customer feedback datasets.
- **NumPy:** Used for numerical operations and data processing.
- **NLTK / spaCy:** Used for Natural Language Processing and text preprocessing.
- **Scikit-learn:** Used for developing and evaluating Machine Learning classification models.
- **TF-IDF / Text Vectorization:** Used to convert textual data into numerical features.

- **Database:** Used to store customer feedback, sentiment results, intent categories, and analysis records.
- **Business Intelligence Tool:** Used to create interactive dashboards, charts, graphs, and reports.
- **Matplotlib / Seaborn:** Used for generating data visualizations during analysis.

These technologies provide the required environment for implementing an efficient customer feedback analysis and Business Intelligence platform.

3.3 Solution Overview

The proposed solution provides an automated platform for understanding customer opinions and intentions from textual feedback. The system accepts customer feedback as input and performs data preprocessing to remove noise and prepare the text for analysis. NLP techniques are then applied to extract useful features from the processed data. The sentiment analysis component determines whether the customer feedback is **positive, negative, or neutral**. The intent analysis component identifies the purpose of the message, such as **complaint, inquiry, purchase request, support request, cancellation, or appreciation**.

The analysis results are stored in a structured database and connected to a Business Intelligence dashboard. The dashboard provides visual representations of customer sentiment, intent distribution, frequently reported issues, and sentiment trends. These insights enable business managers to understand customer behavior, identify areas requiring improvement, and make informed decisions.

Overall System Flow:

Customer Feedback → Data Collection → Data Preprocessing → Feature Extraction → Sentiment Analysis + Intent Analysis → Database → BI Dashboard → Business Insights

3.4 Engineering Standards Applied

The development of the platform follows basic software engineering principles and standards to ensure that the system is reliable, maintainable, scalable, and user-friendly. The system is designed using a **modular architecture**, where data collection, preprocessing, analysis, storage, and visualization are treated as separate components. This makes the system easier to develop, test, maintain, and upgrade.

The following engineering practices are considered:

- **Modularity:** The system is divided into independent functional modules.
- **Maintainability:** Clear and organized code structure is followed for easier modification.
- **Scalability:** The design allows additional datasets, intent categories, and analysis models to be added.
- **Data Validation:** Input data is checked and cleaned before processing.
- **Reliability:** Appropriate error handling and validation are applied during system execution.
- **Security:** Customer information and stored feedback should be protected from unauthorized access.
- **Usability:** Dashboards are designed to present information clearly and allow easy interpretation.
- **Testing:** Individual modules and the complete workflow are tested to ensure correct results.
- **Performance:** Efficient data-processing techniques are used to handle large amounts of customer feedback.

3.5 Solution Justification

The proposed solution is suitable because businesses generate large amounts of customer feedback that cannot be efficiently analyzed through manual methods. A combination of **NLP and Machine Learning** allows the system to automatically understand customer sentiment and intent from textual information. This reduces manual effort and provides faster analysis compared with traditional approaches.

The integration of **Business Intelligence dashboards** further improves the usefulness of the system by converting analysis results into easily understandable visual information. Managers can quickly identify negative sentiment, common complaints, customer requirements, and changing trends. The system therefore supports faster responses, better customer service, improved products and services, and data-driven decision-making.

The solution is also flexible and can be adapted to different business domains such as **e-commerce, banking, retail, telecommunications, hospitality, and customer support**. Its modular design allows new sentiment categories, intent classes, data sources, and machine learning models to be incorporated in the future. Hence, the proposed platform provides a practical, scalable, and intelligent solution for converting customer feedback into valuable business intelligence.

CHAPTER 4
RESULTS AND RECOMMENDATIONS

4.1 Screenshots

The developed **Customer Sentiment and Intent Analysis Platform** provides an interface for analyzing customer feedback and displaying the results in an understandable format. The screenshots of the implemented system can include the **customer feedback input page, data preprocessing output, sentiment analysis results, intent classification results, and Business Intelligence dashboard**. The dashboard displays important information such as the percentage of positive, negative, and neutral feedback, distribution of different customer intents, frequently occurring complaints, and sentiment trends. These visualizations help users understand the analyzed customer data quickly and effectively.

Suggested screenshots to include:

- Customer feedback input screen
- Data preprocessing/output screen
- Sentiment classification results
- Intent classification results
- Business Intelligence dashboard
- Sentiment and intent visualization charts

4.2 Evaluation of Results

The performance of the proposed system is evaluated based on its ability to correctly classify customer sentiment and intent. The system processes the given customer feedback and generates appropriate sentiment and intent categories. The results can be evaluated using metrics such as **accuracy, precision, recall, and F1-score**. The Business Intelligence dashboard is also evaluated based on the clarity and usefulness of its visualizations. The analysis demonstrates that the platform can efficiently transform customer feedback into meaningful information and provide useful insights for business decision-making.

4.3 Challenges Faced

During the development of the platform, several challenges may be encountered. Customer feedback can contain spelling mistakes, abbreviations, emojis, incomplete sentences, and mixed languages, which can affect NLP processing. Another challenge is that the same word or sentence may have different meanings depending on its context. Correctly identifying customer intent can also be difficult when a single message contains multiple requirements. The availability of properly labelled datasets can be another limitation for training Machine Learning models. In addition, handling large volumes of feedback while maintaining good system performance can require efficient data-processing techniques.

4.4 Proposed Improvements

The platform can be improved by using more advanced NLP and Machine Learning models to increase sentiment and intent classification accuracy. The system can be enhanced to support **real-time customer feedback analysis**, allowing businesses to receive immediate alerts about negative feedback or important customer complaints. Additional intent categories can also be introduced to provide more detailed customer insights. The dashboard can be improved by adding advanced filtering, product-wise comparison, time-based analysis, and automated reports. Integration with customer relationship management systems can further improve the practical usefulness of the platform.

4.5 Future Recommendations

In the future, the platform can be extended with **multilingual sentiment and intent analysis** to process customer feedback in different languages. Emotion detection can be added to identify emotions such as happiness, anger, frustration, and satisfaction. Advanced transformer-based models can be implemented to improve contextual understanding of customer messages. The system can also include **predictive analytics** to identify potential customer dissatisfaction and predict future customer behavior. Integration with chatbots and automated customer-support systems can enable immediate responses to customer queries. Real-time data collection from social media and review platforms can further expand the capabilities of the system and provide businesses with continuous customer intelligence.

CHAPTER 5
REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

The development of the Customer Sentiment and Intent Analysis Platform provided valuable knowledge and practical experience in the areas of Natural Language Processing, Machine Learning, Data Analysis, and Business Intelligence. The project helped in understanding how raw customer feedback can be collected, cleaned, processed, and converted into useful information.

Through this project, practical knowledge was gained in Python programming, text preprocessing, feature extraction, sentiment classification, and intent detection. The project also provided experience in working with datasets and understanding how Machine Learning models can be used to solve real-world business problems. In addition, the development of dashboards improved knowledge of data visualization and helped in understanding how analytical results can be presented clearly to business users.

The project also improved problem-solving, analytical thinking, programming, documentation, teamwork, and project management skills.

5.2 Challenges Encountered and Overcome

Several challenges were encountered during the development of the project. One major challenge was handling unstructured customer feedback containing spelling mistakes, special characters, abbreviations, and incomplete sentences. This was addressed by applying appropriate text preprocessing and normalization techniques.

Another challenge was correctly identifying sentiment when customer messages contained mixed opinions or context-dependent expressions. Selecting suitable features and classification techniques helped improve the analysis. Identifying customer intent was also challenging because different customers may use different words to express the same requirement.

Managing and presenting large amounts of analysis results was another challenge. This was addressed by organizing the results into meaningful categories and using Business

Intelligence dashboards for visualization. These challenges helped improve technical knowledge and the ability to solve practical problems systematically.

5.3 Applications of Engineering Standards

Engineering standards and software development principles were considered throughout the project to improve the quality, reliability, and maintainability of the system. The platform follows a modular development approach, where data processing, sentiment analysis, intent classification, storage, and visualization are treated as separate components.

Proper data validation and preprocessing techniques were applied to ensure that the input data was suitable for analysis. The system design also considers reliability, scalability, usability, security, and maintainability. Testing was performed on different stages of the system to identify errors and verify the correctness of the results.

Following these engineering practices helped in developing a structured and dependable platform that can be further enhanced with additional features and technologies.

5.4 Industry Insights

The project provided an understanding of how organizations use customer data and Artificial Intelligence to improve business operations. In real-world industries, companies receive customer feedback from multiple digital channels and require automated systems to analyze this information efficiently.

The project demonstrated the importance of combining NLP, Machine Learning, and Business Intelligence for customer analytics. Sentiment analysis can help organizations monitor customer satisfaction, while intent analysis can help customer-support teams understand and respond to customer requirements. Business Intelligence dashboards can provide managers with a quick overview of important trends and business performance.

The project also provided insight into the importance of data quality, model accuracy, visualization, privacy, and continuous system improvement in developing real-world AI-based business applications.

5.5 Conclusion of Personal Development

The development of this project contributed significantly to both technical and personal development. It provided practical experience in applying theoretical concepts of NLP, Machine Learning, data analytics, and Business Intelligence to a real-world problem. The project improved programming skills and developed a better understanding of the complete process of designing and implementing an intelligent data-analysis system.

It also improved problem-solving, critical thinking, communication, time management, and documentation skills. Working through different technical challenges increased confidence in developing software solutions and encouraged the exploration of advanced AI technologies.

Overall, the project provided valuable practical knowledge and a strong foundation for future work in Artificial Intelligence, Data Science, Natural Language Processing, and Business Intelligence.

CHAPTER 6
CONCLUSION

The Customer Sentiment and Intent Analysis Platform for Business Intelligence successfully provides an intelligent and automated approach for understanding customer feedback. In the present digital business environment, customers communicate their experiences and opinions through various platforms such as product reviews, surveys, social media, emails, chat messages, and customer support systems. The large volume of information generated through these channels makes manual analysis difficult, time-consuming, and less efficient. The proposed system addresses this problem by applying Natural Language Processing, Machine Learning, and Business Intelligence techniques to automatically process and analyze customer-generated text.

The platform performs several important operations, beginning with the collection and preprocessing of customer feedback. The text data is cleaned and transformed into a suitable format using NLP techniques such as tokenization, normalization, stop-word removal, and feature extraction. The processed data is then analyzed to determine the sentiment expressed by customers. The system categorizes feedback into positive, negative, and neutral sentiments, helping organizations understand the overall satisfaction level of their customers.

In addition to sentiment analysis, the platform performs customer intent analysis. This provides a deeper understanding of what customers actually want or expect from the organization. Customer messages can be categorized into intents such as complaints, product inquiries, purchase interest, support requests, feedback, cancellation requests, refund-related requests, and appreciation. Identifying these intentions helps organizations respond to customer requirements more efficiently and assign issues to the appropriate departments.

CHAPTER 7
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CHAPTER 8
APPENDICES

APPENDIX A– DATA PREPROCESSING

The customer feedback collected by the system may contain unwanted characters, duplicate information, unnecessary words, and spelling variations. Therefore, preprocessing is performed before applying sentiment and intent analysis.

The major preprocessing steps include:

1. **Data Cleaning** – Removing unnecessary symbols, special characters, and unwanted information.
2. **Lowercase Conversion** – Converting all text into a standard lowercase format.
3. **Tokenization** – Dividing sentences into individual words or tokens.
4. **Stop-word Removal** – Removing frequently occurring words that provide limited analytical value.
5. **Normalization** – Converting different forms of words into a standard representation.
6. **Duplicate Removal** – Identifying and removing repeated feedback records.
7. **Feature Extraction** – Converting processed text into numerical features suitable for Machine Learning models.

The preprocessing stage improves the quality of customer feedback and helps the analysis model produce more consistent results.

APPENDIX B – SAMPLE PYTHON CODE

The following type of Python implementation can be used for basic customer sentiment analysis:

```
import pandas as pd

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score


# Load dataset

data = pd.read_csv("customer_feedback.csv")


# Input and output

X = data["Feedback"]
```

```
y = data["Sentiment"]

# Convert text into numerical features
vectorizer = TfidfVectorizer(stop_words="english")
X_vectorized = vectorizer.fit_transform(X)

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X_vectorized, y, test_size=0.2, random_state=42
)

# Train model
model = LogisticRegression()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Model Accuracy:", accuracy)
```

The code demonstrates the basic process of converting customer feedback into numerical features, training a classification model, generating predictions, and evaluating the model.

APPENDIX C – SAMPLE INTENT CLASSIFICATION

The platform can categorize customer messages into different intent classes.

Intent Category	Example Customer Message
Complaint	“The product I received is damaged.”
Product Inquiry	“What are the features of this product?”
Purchase Interest	“I am interested in buying this product.”
Support Request	“I need help with my account.”
Cancellation	“I want to cancel my order.”
Refund Request	“Please provide a refund for my order.”
Feedback	“The application is easy to use.”
Appreciation	“Thank you for providing excellent service.”

Intent classification helps businesses understand the purpose behind customer communication and take suitable actions.

APPENDIX D – SAMPLE SENTIMENT ANALYSIS OUTPUT

The following table represents sample results generated by the proposed system:

Customer Feedback	Predicted Sentiment	Predicted Intent
“Excellent service and fast delivery.”	Positive	Appreciation
“The product arrived damaged.”	Negative	Complaint
“What is the price of this product?”	Neutral	Product Inquiry
“I want to buy this product.”	Positive	Purchase Interest
“I need help with my order.”	Neutral	Support Request
“I am very unhappy with the service.”	Negative	Complaint

These results demonstrate how the platform can simultaneously identify the emotional sentiment and purpose of customer feedback.

APPENDIX E – BUSINESS INTELLIGENCE DASHBOARD

The Business Intelligence dashboard provides a graphical representation of the analyzed customer data. The dashboard can contain:

- Overall Sentiment Distribution
- Positive, Negative, and Neutral Feedback Percentage
- Customer Intent Distribution
- Product-wise Sentiment Analysis
- Monthly or Weekly Sentiment Trends
- Frequently Reported Customer Complaints
- Customer Satisfaction Indicators
- Most Common Customer Requests
- Positive and Negative Feedback Comparison
- Key Business Insights and Recommendations

The dashboard allows managers and business analysts to understand large amounts of customer information quickly without manually reviewing every feedback record.